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Network Congestion Analysis in Campus WiFi Systems

Data-Driven Analysis of Network Congestion in Campus WiFi Systems for Performance Optimization

Key Insights

1. Higher number of devices → **lower network speed & performance**
2. Peak usage at **evening/night** → **maximum congestion**
3. Performance varies across **departments & hostels**
4. Strong signal → **better speed & stability**
5. Network quality depends on **multiple factors combined**
6. Data shows **high variability** → **inconsistent user experience**
7. Clear clusters of **high vs medium vs low performing areas**
8. Top performers = **balanced load + strong signal**
9. Bottom performers = **overcrowding + weak signal**
10. **Device density** is the most critical factor affecting **network performance**
11. **Peak hours** amplify **congestion** significantly
12. **Signal strength** alone does not guarantee high speed
13. **Latency** and **packet loss** are strong indicators of congestion
14. Some departments/hostels are **structurally overloaded**, not just temporarily congested